CECH COHOMOLOGY COMPUTES COHOMOLOGY WHEN THE COVERING HAS VANISHING HIGHER COHOMOLOGY ON ALL INTERSECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the hypothesis is translated into the vanishing of the off-row E_2 terms and degeneration is invoked for that situation.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cech-to-cohomology spectral sequence of a covering. Same chapter, the spectral sequence with E_2^{p,q} = H^p(U, underline H^q(F)) converging to H^{p+q}(U, F). Supplies the spectral sequence whose degeneration at E_2 gives the result.
- Cech cohomology of a covering with coefficients in an abelian sheaf. Same chapter, the Cech complex of a covering and its cohomology. Supplies the left hand side of the asserted identification, which is the E_2 row q = 0.
- Degeneration of a first quadrant spectral sequence at E_2. Same chapter, the standard consequence that a spectral sequence whose E_2 terms vanish off one row degenerates, identifying that row with the abutment. Supplies the step from the vanishing of the E_2 terms off q = 0 to the identification of the abutment with that row.

Source reference. Original lemma 03AZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1475–1475. Same-chapter navigation pointer invoked at proof lines 1516-1517, where the author uses the Cech-to-cohomology spectral sequence. The pointer only resolves that reference for the reader; the cited spectral sequence is not reproduced inside the counted span, and no bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a covering. Let $\mathcal{F} \in \mathrm{Ob}(Ab(\mathcal{C}))$. Assume that $H^i(U_{i_0} \times_U \ldots \times_U U_{i_p}, \mathcal{F}) = 0$ for all $i > 0$, all $p \geq 0$ and all $i_0, \ldots, i_p \in I$. Then $\check{H}^p(\mathcal{U}, \mathcal{F}) = H^p(U, \mathcal{F})$.*

Proof. We will use the spectral sequence of Lemma 03AZ. The assumptions mean that $E_2^{p,q} = 0$ for all (p, q) with $q \neq 0$. Hence the spectral sequence degenerates at E_2 and the result follows. □